

NORTON UNIVERSITY

GRADUATE SCHOOL

By: SEK SOCHEAT (June, 2026)



Machine Learning Course Midterm and Final Group Assignment Guideline

For proposal development, proposal defense, final project implementation, and international-journal-style paper writing

Course	Machine Learning
Group Size	Maximum 4 students per group
Programming Language	Any language may be used, but a Machine Learning algorithm is required
Final Goal	A complete ML project with code, evaluation, final defense, and standard academic paper

1. Purpose of the Assignment

This guideline is designed to help students complete both the midterm and final assignments in a clear, professional, and academically standard way.

Students will work in groups to select a Machine Learning topic, prepare a proposal, defend the proposal, build the project, and submit a final paper in international-journal-style format.

The assignment is designed to develop technical skills, research skills, academic writing skills, teamwork, and presentation skills.

2. General Rules

- Students must work in groups of no more than 4 members.

- Each group must choose one topic only.
- Each topic must be related to a real Machine Learning problem.
- Any programming language or development platform is allowed.
- The project must include at least one Machine Learning algorithm.
- The project must include data, preprocessing, model training, evaluation, and result interpretation.
- Each group must complete both the midterm and final submission requirements.

3. Topic Selection

Students may choose any feasible ML topic. Example areas include:

- House price prediction
- Student performance prediction
- Student dropout risk prediction
- Customer segmentation
- Disease prediction
- Spam detection
- Sentiment analysis
- Product recommendation
- Loan approval prediction
- Image classification

The lecturer may approve or revise a topic if it is too weak, too broad, too simple, too difficult, or not clearly related to ML.

4. Required ML Components

- Problem statement
- Dataset or data source
- Feature selection or feature engineering
- Data cleaning and preprocessing
- Model training and testing
- At least one Machine Learning algorithm
- Performance evaluation using suitable metrics

- Interpretation of results
- Practical conclusion

5. Midterm Assignment: Proposal Development and Defense

At midterm, each group must submit a proposal and defend it orally. The proposal must show that the project is meaningful, practical, and academically sound.

5.1 Required Proposal Contents

1. Project title
2. Group members and roles
3. Background and problem statement
4. Objectives of the project
5. Research questions or expected outcomes
6. Target users or beneficiaries
7. Dataset source or data collection plan
8. Features and target variable
9. Proposed ML algorithm(s)
10. Proposed evaluation metrics
11. Programming language, framework, and tools
12. Expected outputs of the system
13. Work plan and timeline
14. Possible limitations or risks

5.2 Proposal Defense Guidance

- Recommended presentation time: 8 to 12 minutes
- Recommended question-and-answer time: 5 to 10 minutes
- All group members should understand the full project
- The group should explain why the topic matters, what data will be used, why the selected algorithm is suitable, and how success will be measured

5.3 Suggested Midterm Rubric

Criterion	Description	Suggested Weight
Topic quality	Relevant, meaningful, and clearly related to ML	15%

Problem clarity	The problem and objectives are clearly explained	15%
Data plan	The data source or collection plan is realistic	15%
Methodology	Features, algorithms, and metrics are suitable	20%
Feasibility	The work plan is realistic	10%
Proposal writing	Clear, organized, and professional proposal	10%
Defense	Clear presentation and strong answers	15%

6. Final Assignment: Complete Project and Journal-Style Paper

At the end of the course, each group must submit a complete Machine Learning project and a full academic paper written in a standard international-journal-style structure.

6.1 Final Deliverables

- Final source code
- Dataset or clear dataset reference
- Project paper in DOCX or PDF
- Presentation slides
- System demonstration or output demonstration
- Any needed appendix or supplementary material

6.2 Final Defense Guidance

- Recommended presentation time: 12 to 15 minutes
- Recommended question-and-answer time: 8 to 12 minutes
- The group must explain both the implementation and the paper
- Students must explain the dataset, preprocessing, algorithm selection, evaluation metrics, results, limitations, and future work

7. Standard Paper Structure

The final paper should follow a clean academic structure similar to an international journal paper.

Title

Author names and affiliations

Abstract

Keywords

1. Introduction

2. Literature Review or Related Work

3. Materials and Methods / Methodology

4. Results

5. Discussion

6. Conclusion and Future Work

References

7.1 Title

The title should be specific, academic, and clearly describe the project topic.

Example: "Machine Learning-Based Student Dropout Prediction Using Logistic Regression and Random Forest"

7.2 Abstract

The abstract should usually be 150 to 250 words and should briefly include:

- Background or problem
- Objective
- Dataset or method
- Main algorithm(s)
- Important result(s)
- Main conclusion

7.3 Keywords

Provide 4 to 6 keywords, for example: Machine Learning, Classification, Prediction, Dataset, Model Evaluation.

7.4 Introduction

- Explain the real problem
- Explain why the problem is important
- State the objective of the study
- State the contribution of the project

7.5 Literature Review / Related Work

This section should summarize previous studies, similar systems, or relevant ML approaches. Students should compare their work with previous work and explain what is similar or different.

7.6 Materials and Methods / Methodology

- Dataset description
- Data source or collection method
- Number of records and attributes
- Features and target variable
- Missing value handling
- Encoding or scaling method
- Train-test split or cross-validation
- Selected ML algorithm(s)
- Evaluation metrics
- Software tools and implementation method

7.7 Results

This section should show clear evidence, not only general statements.

- Performance metrics such as accuracy, precision, recall, F1-score, MAE, RMSE, R2, silhouette score, or other suitable metrics
- Model comparison tables
- Confusion matrix for classification if relevant
- Graphs or charts for comparison and trends
- Screenshots or output figures if needed

7.8 Discussion

- Explain why the model performed well or poorly
- Explain what features may have influenced the results
- Discuss limitations
- Compare results with previous work if possible
- Explain the practical meaning of the findings

7.9 Conclusion and Future Work

The conclusion should summarize the problem, method, key findings, and contribution. Future work should explain how the project can be improved or extended.

7.10 References

Use a consistent citation style. The lecturer may require IEEE style. Students must not mix citation styles in one paper.

8. Guidance for Figures, Tables, Graphs, and Charts

8.1 Figures

- Use figures for workflow diagrams, architecture diagrams, screenshots, sample outputs, or model illustrations
- Every figure must have a figure number and caption
- The figure should be discussed in the text
- Figures must be clear and readable

Example: "Figure 2 shows the overall workflow of the proposed system."

8.2 Tables

- Use tables for dataset description, feature lists, parameter settings, and model comparison
- Every table must have a table number and caption
- Keep tables clear and not overcrowded
- Use consistent units and decimal places

Example: "Table 3 presents the comparison of model performance."

8.3 Graphs and Charts

- Use bar charts for category comparison
- Use line charts for trends over time
- Use pie charts only for simple part-to-whole data
- Use scatter plots for relationships between variables
- Always label axes and include units when needed
- Explain each graph or chart in the text

8.4 Placement Rule

Figures, tables, and charts should be placed as close as possible to the paragraph where they are explained.

9. Suggested Formatting Standard

Item	Recommended Format
Paper length	Usually 8 to 15 pages, depending on project quality
Font	Times New Roman or Arial
Font size	11 or 12 pt for body text
Line spacing	1.15 or 1.5
Margins	Normal academic margins such as 1 inch or 2.54 cm
Alignment	Justified body text is recommended
Page numbering	Include page numbers unless the template says otherwise

10. Required Evidence in the Final Paper

- A clear dataset description
- At least one methodology or workflow figure
- At least one table describing features, dataset, or parameters
- At least one evaluation table comparing results
- At least one graph or chart if appropriate
- A discussion of limitations and future work

11. Suggested Student Workflow

1. Form a group and assign roles
2. Choose a topic and get the lecturer's approval
3. Prepare and defend the midterm proposal
4. Collect or prepare the data

5. Clean and preprocess the data
6. Train and evaluate at least one ML model
7. Improve the project if needed
8. Write the paper section by section
9. Insert figures, tables, charts, and references correctly
10. Prepare the final presentation and defense

12. Group Role Suggestions

- Project leader or coordinator
- Data collection and preprocessing member
- Model development and evaluation member
- Paper writing and formatting member

Even with role division, all members must understand the full project because each member may be questioned in the defense.

13. Academic Integrity and Ethics

- Do not copy another group's code or paper
- Do not copy text without citation
- Do not fabricate data or results
- If AI tools are used, students must still verify, revise, and understand all content
- Respect privacy and ethics when working with human-related data

14. Suggested Final Rubric

Criterion	Description	Suggested Weight
Topic relevance	The project is meaningful and suitable for ML	10%
Dataset and preprocessing	Data is relevant, sufficient, and well prepared	15%
ML methodology	Algorithms and evaluation are appropriate	20%
Implementation	The system or model works correctly	15%

Results and interpretation	Results are clearly presented and explained	15%
Paper quality	Paper follows academic structure and standard formatting	15%
Final defense	Presentation and answers are clear and professional	10%

15. Student Submission Checklist

- ☐ My group has no more than 4 members
- ☐ My topic is approved
- ☐ My project includes at least one ML algorithm
- ☐ My code runs and produces results
- ☐ My paper includes standard academic sections
- ☐ My paper includes figures, tables, and charts where appropriate
- ☐ All visuals have captions and are explained in the text
- ☐ My references are cited consistently
- ☐ All group members are ready for the defense

16. Final Advice

- Choose a project that is meaningful, manageable, and supported by data.
- Do not wait until the end to write the paper. Write continuously during project development.
- Use figures, tables, graphs, and charts to support your findings, not only for decoration.
- A strong Machine Learning project is not only about high performance. It is also about correct methodology, clear explanation, and useful interpretation.